Fleet Progress Report

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CSCE552 Game Development, Dr. JJ Shepherd

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I. General Progress Update

Summer camps at GSSM have now ended and I am done with the AI in Education conference that I was a speaker at. *I have no other obligations the next two weeks except making the game for this course.* I anticipate that I will still accomplish all necessary tasks by their deadlines. A list of tasks I did this week is below. Project commit history at https://github.com/hotdogontology/CSCE552/commits/main/

1) Completed tutorial on GDScript.
2) Completed tutorial on making 2D games in Godot.
3) Made stand-in ship model in Blockbench.
4) Started and abandoned tutorial on making 3D games in Godot in favor of beginning work on game.

II. Features Currently Implemented

I have been able to create a basic 3D scene, import my model, and have it move around with the camera in approximately the correct place. It is very much NOT a game right now, but I am happy with my progress given that I have never made a 3D game before or used Godot and my other obligations so far.

1) A basic scene tree has been established in the project. Current scenes are
   a) Main
   b) Player
   c) Trench
   d) TrenchScraper (Building)
2) A player ship is drawn within a trench scene.
3) The player can move the ship in four directions.

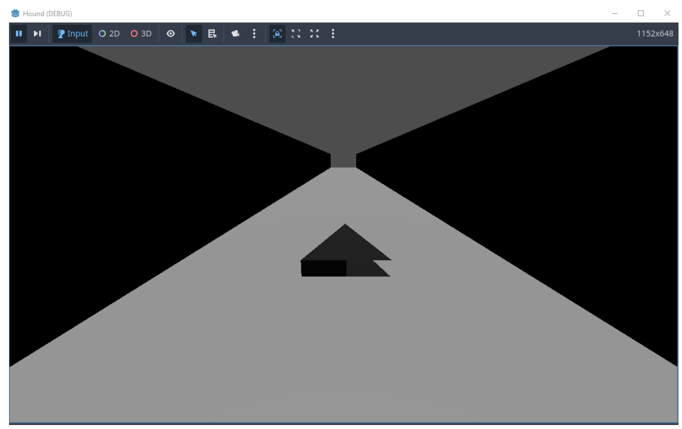

Fig. 1. screenshot of current game

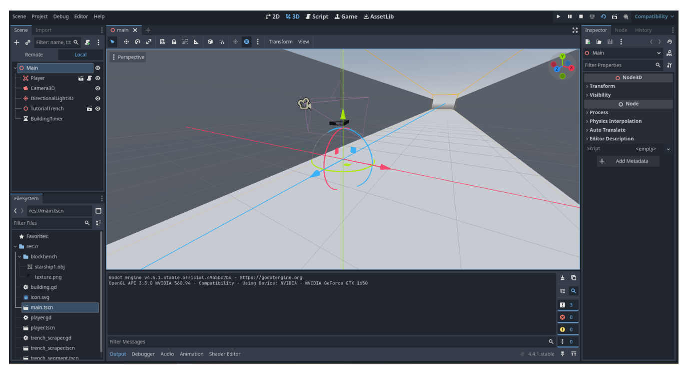

Fig. 2. current 3d editor view of main scene

III. Features To Be Implemented

All other features as described in the GDD submitted last week. Essentially, now that I understand Godot a little, I need to actually make the game.

1) Smoother player movement that feels like a ship flying
2) Shooting weapons
3) Collisions with objects and enemy weapons
4) Health and shield systems
5) Loadout scene and systems
6) Enemy and obstacle spawn and movement
7) Random level generation and on-rail levels
8) Galaxy map scene
9) All menus

IV. Features Removed

I still plan to make the game as described in the GDD to at least the minimum specifications.

V. Problems Encountered

1) The largest obstacle for me this past week was time. I had obligations for the AI conference from 9 AM - 5 PM Monday - Thursday and had to work in the evenings. **But now I have the next two weeks completely free.**
2) Audacity is audio-editing software, but as far as I can tell, it cannot allow me to write music. I did not realize this. I will need to download additional tracking software. I am looking into Reaper and NES extensions to write some chiptune music as well as investigating if I can export sounds or music from PICO-8's built-in tracker.
3) My triangle spaceship model is missing coloring on some faces and I am not sure why. It was my first blockbench model ever, so I probably just colored the mesh incorrectly. The triangle model is a placeholder for the final model, so this is mainly a non-issue.
4) Related to the above, making 3D art in Blockbench is very difficult! But I am sticking to basic shapes to manage.
5) Godot makes scene management and built-in features as described in lecture very manageable, but I am finding that style of organization for writing a game far more un-intuitive than coding everything like I have in PICO-8 games. My sense of "*here is what code you should write to implement that feature*" that I have developed is often unrelated to the new sense of "*here is the node type that you should create to implement that feature and what characteristics you should edit*" that Godot requires. That is, the learning curve is steeper than I anticipated. However, the tutorials helped and I am getting the hang of it.
6) I can get the player to move up, down, left, right, but the movement does not *feel* like a spaceship moving yet. I need to implement the nose tipping up and down and the ship banking. I also need to have the camera probably move slightly with the ship. I will need to do some research into how to get a better game feel for player controls.

VI. Team Tasks

- **Taylor, July 17–20:** Improve player movement, add collisions, enemies, obstacles, weapons, and loadout system.
- **Taylor, July 20–22:** Create MWE in trench tutorial scene with randomly generated enemies. Create at least one on-rails hand-crafted level to test the PathFollow node.
- **Taylor, July 22–24:** Create set-up for all randomly generated scenes, create galaxy map and mission selection system.
- **Taylor, July 24–29:** Create as many on-rails levels as possible.
- **Taylor, July 29–31:** Finishing details, playtest and revise, submit game.
- **Clayton, July 20–31:** Create art and music as necessary to assist Taylor. Playtest and give feedback.

VII. Programming Tasks

- **Taylor, July 17–20:** Improved player movement, collisions signals, enemy movement and behavior, build loadout system.
- **Taylor, July 20–22:** Generating enemies and obstacles for randomly generated levels. Figure out how the path follow node interacts with player input.
- **Taylor, July 22–24:** Mission select and galaxy map logic.
- **Taylor, July 24–29:** Create as many on-rails levels as possible.
- **Taylor, July 29–31:** Finishing details, playtest and revise, submit game.

VIII. Art Tasks

Minimum art assets we need to create. These are all assigned to Taylor by default. Clayton will assist where he can.

- **Taylor, July 17–22:**
  - Enemy Spaceship 1 Model
  - Enemy Spaceship 2 Model
  - Enemy Spaceship 3 Model
  - Enemy Turret 1 Model
  - Laser
  - Bomb
  - Building 1 Model
  - Asteroid Model
  - Asteroid Field Icon
- **Taylor, July 22–24:**
  - Space Station Model
  - Space Station Icon
  - Planet Icon
  - Planet Background
  - Improved Player Spaceship Model
  - Carrier Ship Model
  - Battery Loadout Icon
  - Shield Generator Loadout Icon
  - Missile Loadout Icon
  - Loadout Bay Puzzle
  - Galaxy Map

IX. Testing Tasks

- **Taylor, July 17–20:** Test movement, firing, collisions, energy, shields, armor, pickups, and loadout validation as they are implemented.
- **Taylor, July 20–22:** Test all scene transitions and verify that fleet and campaign data persist.
- **Taylor, July 22–24:** Test victory, failure, retreat, replacement-fighter, save, load, and game-end conditions.
- **Taylor, July 24–29:** Conduct playtests focused on controls, objective clarity, interface readability, mission length, and difficulty.
- **Taylor, July 29–31:** Perform a final test and correct crashes, soft locks, and progression blockers.
- **Clayton, July 20–31:** Conduct playtests focused on controls, objective clarity, interface readability, mission length, and difficulty.